Tien-Chen Lin, Ph.D.

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My research focus lies at the intersection of molecular neurobiology, cytoskeletal dynamics, and mechanotransduction. Specifically, I aim to explore how diverse cell types coordinate their morphological changes during brain development. To achieve this, my research approach integrates theoretical modeling of cytoskeletal dynamics with experimental observations. My strategies will leverage a combination of live imaging and optogenetic manipulation of biomolecules both *in vitro* and *in vivo*, alongside biochemical reconstitutions of cytoskeletal systems. Additionally, I'll harness extensive omics data to prioritize specific biomolecules and pathways for investigation. Ultimately, my long-term objective is to unveil the underlying biophysical principles governing the self-organization of the nervous system. This research holds promise in shedding light on the fundamental processes driving coordinated morphological changes in developing neural tissues and the dysregulation in aging and degenerated brains.

Education

Ph.D., Natural Sciences (Magna cum laude, Dr. rer. nat) **2009–2014**

University of Heidelberg, Germany

M.S., Biochemistry and Molecular Biology **2005–2006**

National Yang-Ming University, Taiwan

B.S., Faculty of Life Science **2002–2005**

National Yang-Ming University, Taiwan

Research experience

Postdoctoral fellow **2017–present**

German Center of Neurodegenerative Disease (DZNE)

Supervisor: Dr. Frank Bradke

- Discovered the mechanical local-excitation global-inhibition system mediated by Arp2/3-actomyosin reciprocal antagonism as the principal mechanism of CNS neuronal polarization.
- Reconciled the contradictory roles of actin networks and clarified the role of CNS growth cones in neurite growth.
- Established the analysis pipeline in R and ImageJ scripts to quantitatively describe neurite growth dynamics.
- Revealed the correlation between the downregulation of pre-synaptic vesicle related genes and the increase of axon regeneration potential (collaboration with Dr. Brett Hilton, University of British Columbia, Canada).
- Revealed the correlation between ischemic stroke and the upregulation of extracellular matrix-related genes in the reactive astrocytes (collaboration with Dr. Gabor C. Petzold, DZNE).
- Revealed the correlations among dorsal root ganglion neuron subtypes, their responses to peripheral nerve lesion and the expression of regeneration-associated genes (collaboration with Dr, Andreas Husch, DZNE).

Postdoctoral fellow **2014–2017**

Zentrum für Molekulare Biologie Heidelberg (ZMBH), University of Heidelberg

Supervisor: Dr. Elmar Schiebel

- Revealed the evolution-conserved protein MOZART1 as the integral part of the microtubule nucleator assembly unit, γ-tubulin small complex, in the pathogenic yeast (*Candia albicans*) and human cells.
- Elucidated MOZART1 as the factor promoting the polymerization of γ-tubulin small complex into the microtubule nucleation-competent ring complex.
- Established the evolution-conserved physical interactions among MOZART1, γ-tubulin small complex and the complex receptors (pericentrin and CDK5RAP2).

Pre-doctoral fellow **2009–2014**

Zentrum für Molekulare Biologie Heidelberg (ZMBH), University of Heidelberg

Supervisor: Dr. Elmar Schiebel

- Elucidated the phospho-regulation of γ-tubulin small complex and its nuclear receptor (pericentrin) mediated by mitotic kinases and its importance in mitosis.
- Revealed the physical interaction between γ-tubulin small complex and its nuclear receptor.
- Discovered the evolution relationship of divergent eukaryotic γ-tubulin complex targeting factors to microtubule organizing centers.

Honors and awards

1. EMBO Short-term Fellowship for *C. albicans* γ-tubulin small complex and MOZART1 project **2016**
2. LGFG Fellowship (Landesgraduiertenförderung-Programme) "Regulation of Cell Division" **2012**

Talks

1. Microtubules in Neurons, Chiemsee **2023**
2. The Cytoskeleton of Neurons and Glia webinars series, virtual **2023**
3. EMBO Workshop Neural development and neurodegeneration, Taipei **2022**
4. FENS Symposium Barriers and hopes for functional restoration after traumatic CNS injury, Paris **2022**
5. CSHL meeting Molecular Mechanisms of Neuronal Connectivity, New York **2022**
6. DZNE Mini Symposium Information Flow, Bonn **2022**
7. European Cytoskeleton Forum, Hannover **2022**
8. Axon Repair Meeting, virtual **2022**
9. DZNE Science Meeting, Bonn **2021**
10. DKFZ-ZMBH Alliance Young Investigators Seminar, Heidelberg **2016**

Teaching and Service

Associate Faculty Member of F1000 Faculty Opinions **2019-present**

- Comment on “Transcriptional reprogramming of distinct peripheral sensory neuron subtypes after axonal injury” by Renthal W et al., doi:10.3410/f.736924635.793570162
- Comment on “Cell membranes resist flow” by Shi Z et al., doi:10.3410/f.734323511.793564815

University of Bonn

- Coordinator of M.Sc. program “Molecular Cell Biology” EM5 practical module **2023**
- M.Sc. “Molecular Cell Biology” EM5 module, lecturer of Data Visualization with R (4 hours) **2023**
- M.Sc. “Molecular Cell Biology” EM5 module, lecturer of Light Microscopy Basics (2 hours) **2022**
- M.Sc. “Neuroscience”, teacher of Live-cell Imaging session (2 days) **2022**
- M.Sc. “Molecular Cell Biology” EM5 module, lecturer of Light Microscopy Basics (2 hours) **2020**

University of Heidelberg

- M.Sc. practical course “Mechanism and Regulation of Eukaryotic Cell Division” (1 week) **2013**
- M.Sc. lecture “Cell Cycle Checkpoint Regulation” (2 hours) **2012**
- Supervision of research project of M.Sc. student Annalena Meyer **2017**
  Project: Investigation of the GRIP1 domain of γ-tubulin complex subunit GCP3.
- Supervision of research project of M.Sc. student Yvonne Schlosser **2014**
  Project: Investigation of the N-terminal domain of γ-tubulin complex subunit GCP3.

Professional Development

1. Project Management for Scientists, EMBO **2021**
2. Advanced Techniques in Molecular Neuroscience, Cold Spring Harbor Laboratory **2020**
3. Super-Resolution Microscopy, EMBL **2020**
4. Laboratory Leadership for Group Leaders, EMBO **2020**
5. Negotiation for Scientists, EMBO **2020**
6. Analysis and Integration of Transcriptome & Proteome Data, EMBL **2018**